Zhangsong Li

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Education

Peking University 2023–2028 (expected)
Ph.D. student in Probability
Advisors: Jian Ding

Peking University 2019–2023
B.S. in Mathematics

Research interests

Probability, statistical inference, combinatorics, and theoretical computer science.

Articles

Journal Publication

(J1) **A Polynomial Time Iterative Algorithm for Matching Correlated Gaussian Matrices with Non-vanishing Correlation**
Jian Ding and Zhangsong Li
Foundations of Computational Mathematics, 25(4):1287–1344, 2025.

(J2) **Low-Degree Hardness of Detection for Correlated Erdős-Rényi Graphs**
Jian Ding, Hang Du, and Zhangsong Li
Annals of Statistics, 53(5):1833–1856, 2025.

(J3) **A Computational Transition for Detecting Correlated Stochastic Block Models by Low-Degree Polynomials**
Guanyi Chen, Jian Ding, Shuyang Gong, and Zhangsong Li
Annals of Statistics, 54(1):226–251, 2026.

(J4) **A Computational Transition for Detecting Multivariate Shuffled Linear Regression by Low-Degree Polynomials**
Zhangsong Li
IEEE Transactions on Information Theory, 72(4):2444–2456, 2026.

(J5) **Asymptotic Diameter of Preferential Attachment Model**
Hang Du, Shuyang Gong, Zhangsong Li, and Haodong Zhu
Electronic Communications in Probability, 31:1–12, 2026.

(J6) **Robust Random Graph Matching in Dense Graphs via an Approximate Message Passing Type Algorithm**
Zhangsong Li
IEEE Transactions on Information Theory, to appear.

(J7) **A Polynomial-Time Iterative Algorithm for Random Graph Matching with Non-vanishing Correlation**
Jian Ding and Zhangsong Li
Mathematics of Operations Research, to appear.

Conference Publication

(C1) **Robust Random Graph Matching in Gaussian Models via Vector Approximate Message Passing**
Zhangsong Li
Proceedings of 38th COLT, pages 3580–3581. PMLR, 2025.
Conference version of (J6).

(C2) **Algorithmic Contiguity from Low-Degree Conjecture and Applications in Correlated Random Graphs**
Zhangsong Li
Proceedings of 29th APPROX/RANDOM, pages 30:1–30:18. Schloss Dagstuhl Leibniz-Zentrum für Informatik, 2025.

(C3) **Detecting Correlation Efficiently in Very Supercritical Stochastic Block Models: Breaking the Otter's Threshold Barrier**
Guanyi Chen, Jian Ding, Shuyang Gong, and Zhangsong Li
Proceedings of 37th SODA, pages 2743–2759. SIAM, 2026.

(C4) **Detection and Reconstruction of a Random Hypergraph from Noisy Graph Projection**
Shuyang Gong, Zhangsong Li, and Qiheng Xu
Proceedings of 59th ISIT, to appear.

Preprints

- **The Umeyama Algorithm for Matching Correlated Gaussian Geometric Models in the Low-Dimensional Regime**
  Shuyang Gong and Zhangsong Li
  Preprint, *https://arxiv.org/abs/2402.15095*
- **A Smooth Computational Transition in Tensor PCA**
  Zhangsong Li
  Preprint, *https://arxiv.org/abs/2509.09904*
- **Detecting Correlation Efficiently in Stochastic Block Models: Breaking Otter's Threshold in the Entire Supercritical Regime**
  Guanyi Chen, Jian Ding, Shuyang Gong, and Zhangsong Li
  Preprint, *https://arxiv.org/abs/2503.06464*
- **The Algorithmic Phase Transition in Correlated Spiked Models**
  Zhangsong Li
  Preprint, *https://arxiv.org/abs/2511.06040*
- **Improved Computational Lower Bound of Estimation for Multi-Frequency Group Synchronization**
  Zhangsong Li
  Preprint, *https://arxiv.org/abs/2601.20522*
- **Fundamental Limits of Community Detection in Contextual Multi-Layer Stochastic Block Models**
  Shuyang Gong, Dong Huang, Zhangsong Li
  Preprint, *https://arxiv.org/abs/2602.08173*
- **Algorithmic Contiguity from Low-Degree Heuristic II: Predicting Detection-Recovery**

Gaps
Zhangsong Li
Preprint, *https://arxiv.org/abs/2604.17410*

Teaching

Fall 2023	TA, Applied Stochastic Process (Honor)	Peking University
Fall 2024	TA, Advanced Probability Theory	Peking University
Spring 2025	TA, Probability Theory	Peking University

Service

- **Assistant in Undergraduate Research Mentorship:**

  (1) Peking University undergraduate research program 2024-2025

  Student: Chenxu Feng (mentored by Jian Ding)

  Title: "Strong Detection Threshold in Correlated Erdős-Rényi Graphs with Constant Average Degree"

  (2) Peking University undergraduate research program 2025-2026

  Students: Chenxu Feng and Yifan Li (mentored by Jian Ding)

  Title: "Structural Properties of the Geometric Preferential Attachment Model"

- **Journal Reviewing:** *Annals of Applied Probability*, *Bernoulli*, *IEEE Transactions on Information Theory*, *Information and Inference: A Journal of the IMA*

Talks

Peking University Elite Training Program, A Polynomial-Time Iterative Algorithm for Random Graph Matching with Non-vanishing Correlation, June 2023.

Tsinghua Sanya International Mathematics Forum, Low-Degree Hardness of Detection for Correlated Erdős-Rényi Graphs, January 2024.

Tsinghua University Statistics Seminar, Recent Progress on Random Graph Matching Problems, March 2025.

YMSC Probability Seminar, Asymptotic Diameter of Preferential Attachment Model (joint with Shuyang Gong), May 2025.

International Conference on Applied Probability, Robust Random Graph Matching in Gaussian Models via Vector Approximate Message Passing, June 2025.

The 38th Annual Conference on Learning Theory, Robust Random Graph Matching in Gaussian Models via Vector Approximate Message Passing, July 2025.

The 29th International Conference on Randomization and Computation, Algorithmic Contiguity from Low-Degree Conjecture and Applications in Correlated Random Graphs (online), August 2025.

Peking University Sino-Russian Student Mathematical Seminar, The Algorithmic Phase Transition for Correlated Spiked Models, December 2025.